Power Spectrum Estimation Examples: Welch's Method

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Key Concepts

1) Bias is smallest for long windows and increases as the window length decreases.

2) Variance decreases as the number of segments averaged increase.

3) If the overall data length is limited, the number of segments is inversely related to the window length used for each segment.

    a) Best bias but worst variance with one segment (no averaging).

b) Bias increases as variance is reduced by using more segments.

4) Overlap of segments increases the number of segments for fixed window and data lengths, but the segments are dependent so the variance is not reduced proportionally.

Video Screenshots

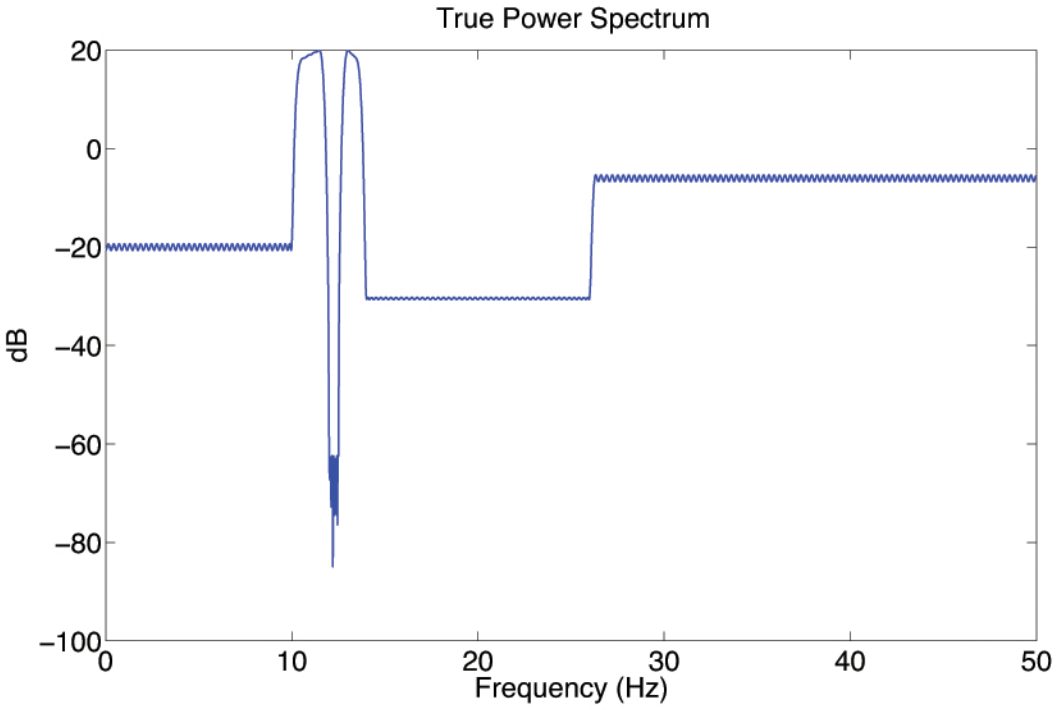
Example 1: Power spectrum using "pwelch"

0:43



$$S_{xx}(\omega) = |H(e^{j\omega})|^2$$

$$S_{xx}(f) = |H(e^{j2\pi f/f_s})|^2$$



pwelch (MATLAB) calculates spectral density ⇒ multiply by $f_s/2$ to get spectrum

$Q = 1024$

$f_s = 100$ Hz

Amplitude units: Watts

Segment length L

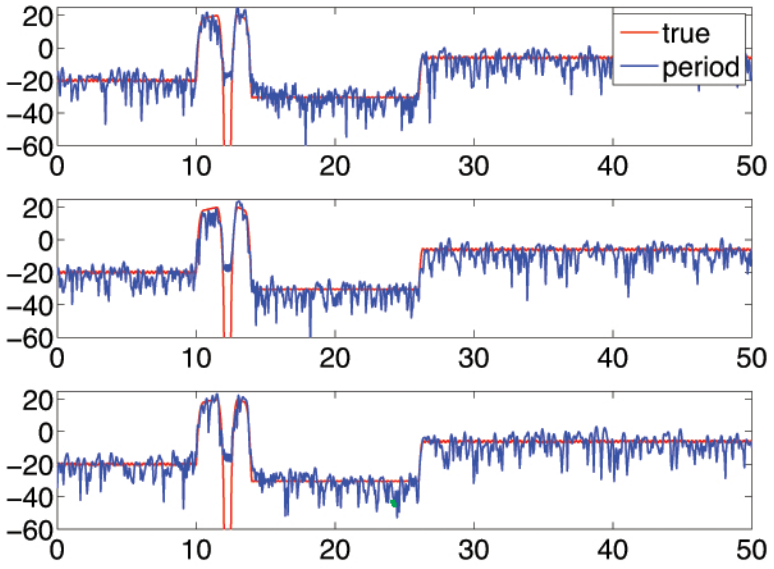
Segment offset R

Example Power Spectrum Estimates in dB

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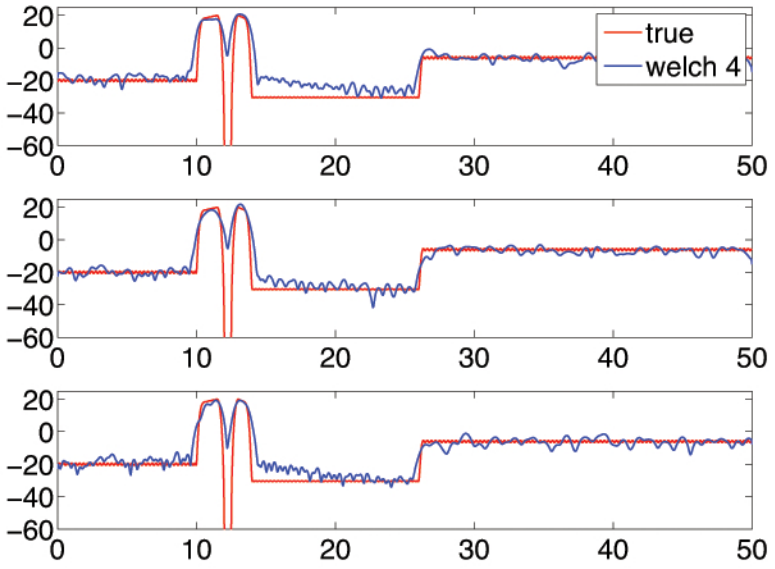
- Hamming window
- No overlap (R = L)
- Effect of averaging on variance

L = 1024, K = 1



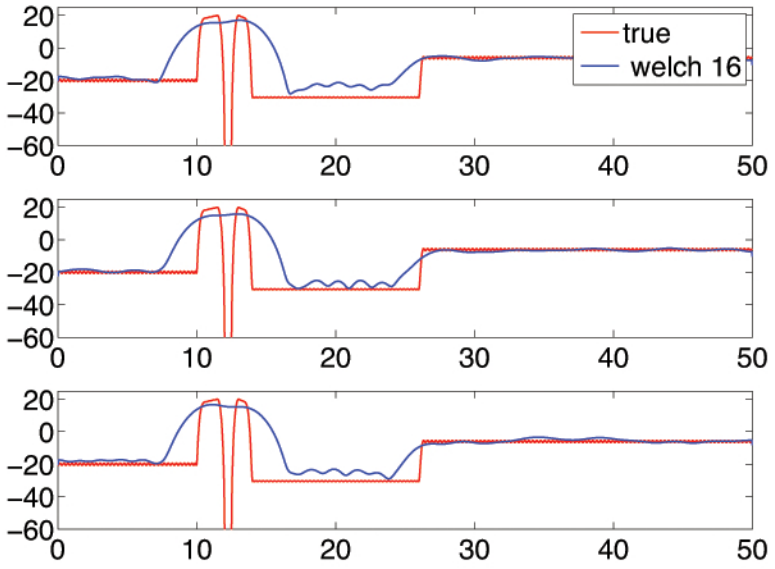
Hz

L = 256, K = 4



Hz

L = 64, K = 16



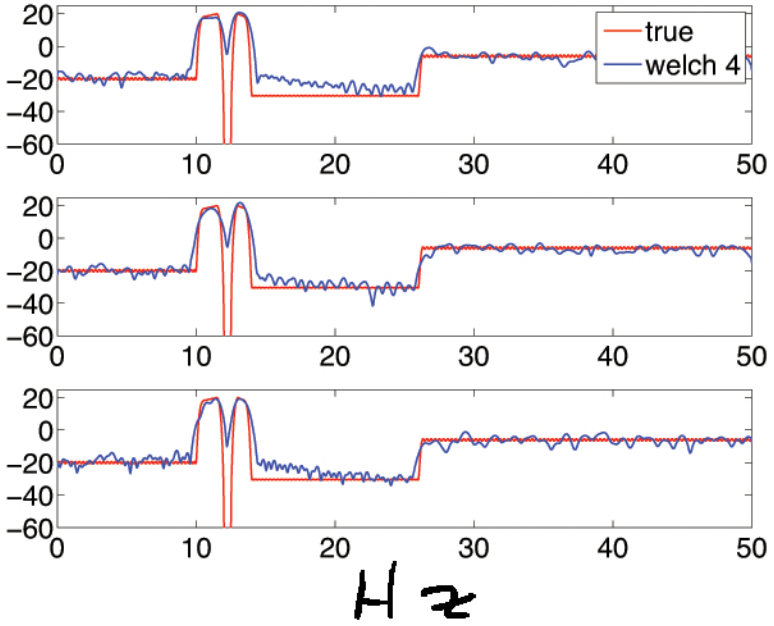
Hz

Example Power Spectrum Estimates in dB

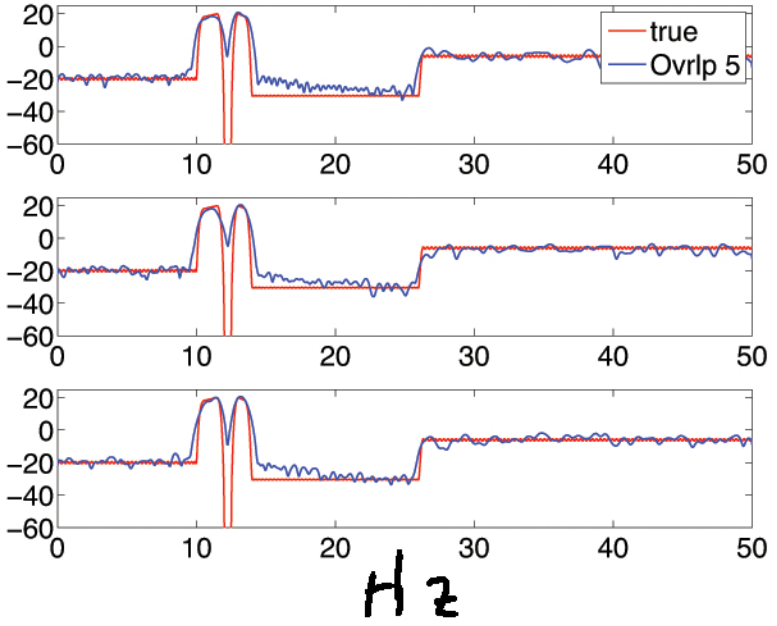
5:47

- Effect of overlap
- Q = 1024, L = 256
- Hamming Window

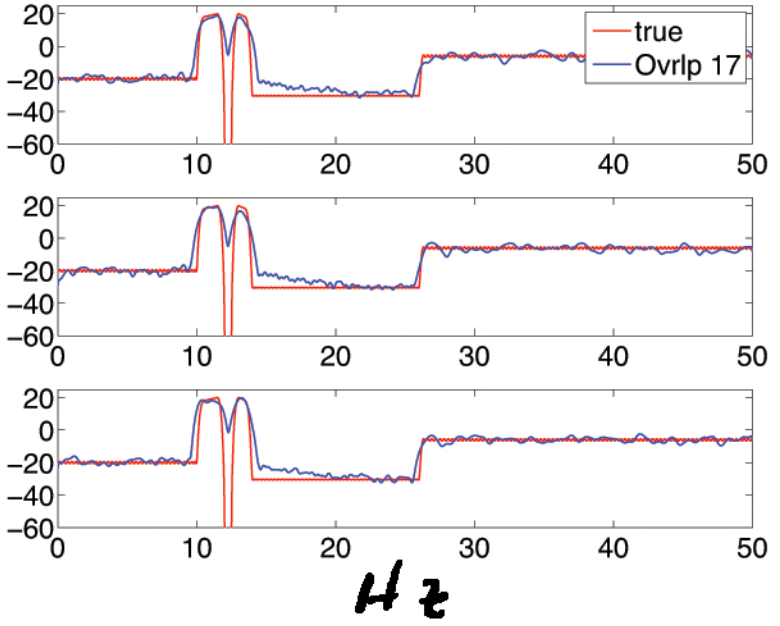
R=256, K=4



R=192, K=5



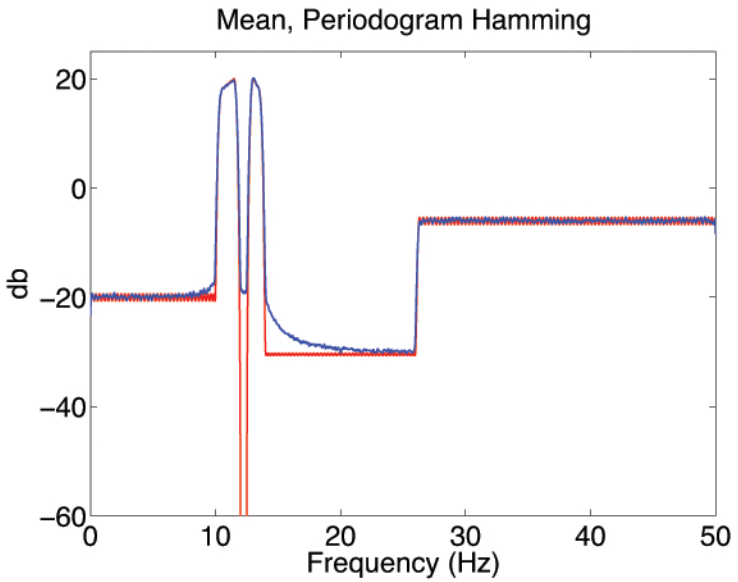
R=48, K=17



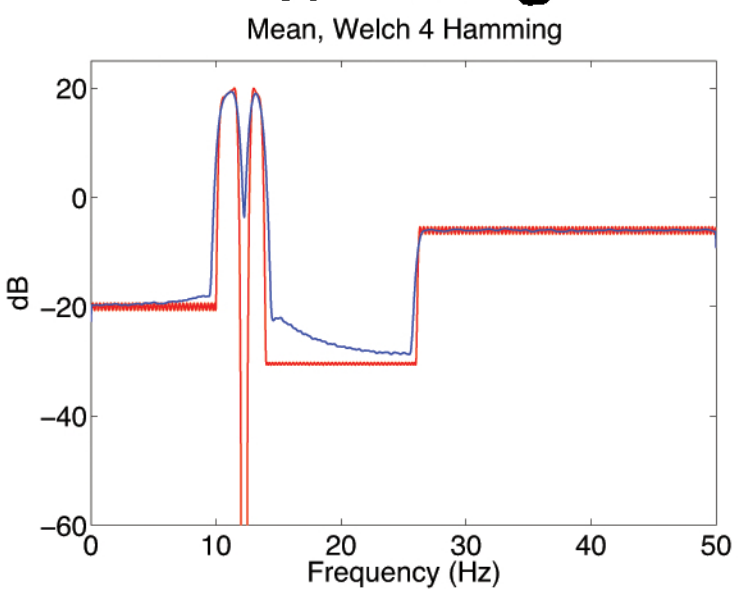
Mean of Power Spectrum Estimate in dB

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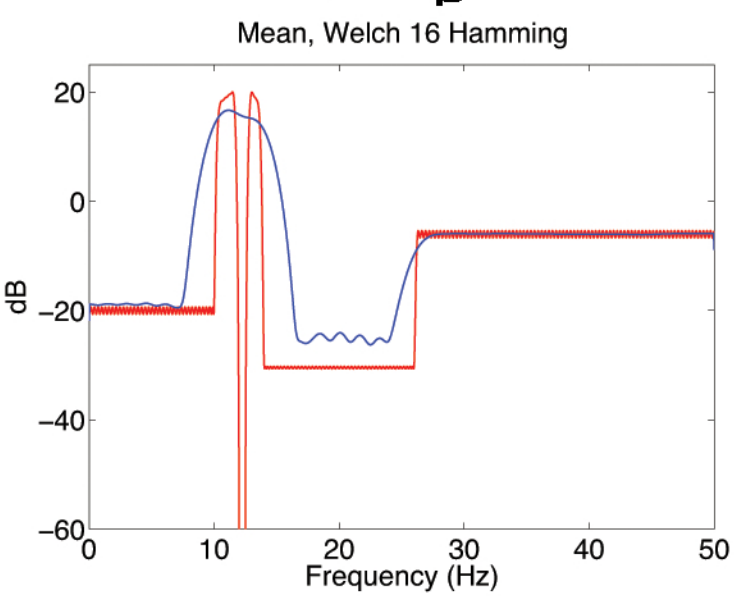
L = 1024

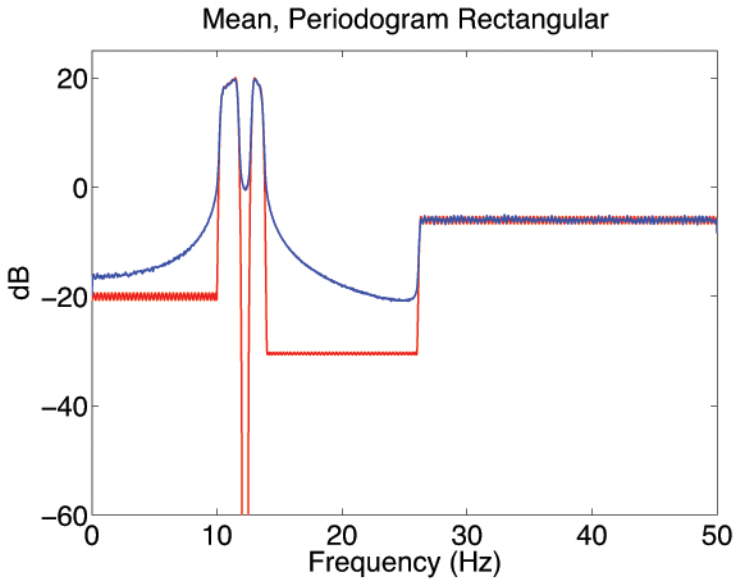


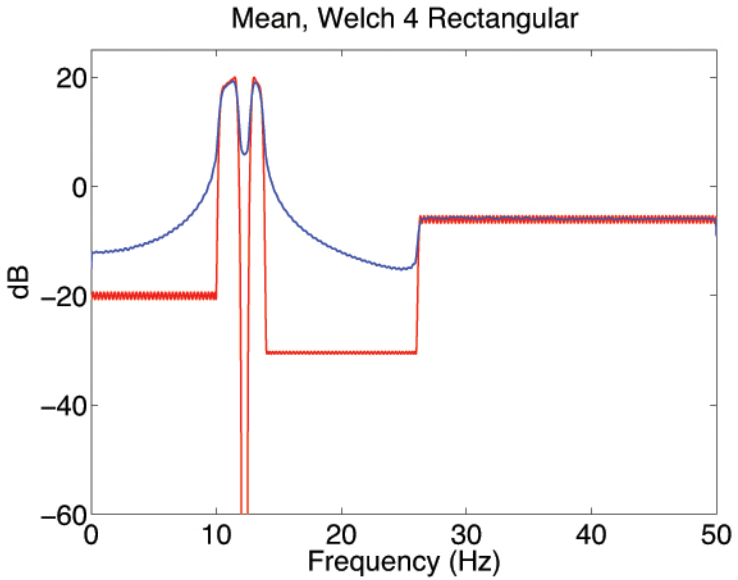
L = 256



L = 64







- Average 500 estimates
- No overlap (R=L)
- Bias:
  - Resolution - mainlobe width
  - Dynamic range - sidelobe height

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